

Sidharth Rao

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EDUCATION

Iowa State University

Ames, IA

Bachelor in Computer Engineering (GPA: 3.4) Dean's List: Fall 2022 - Fall 2023

Aug. 2022 – May 2026

Courses: Data Structures, Design & Analysis of Algorithms, Differential Equations, Linear Algebra, Object-Oriented Programming,, Database Management, Operating Systems: Principles

EXPERIENCE

ARA Wireless Living Lab

May 2025 – August 2025

Undergraduate Research Assistant — Iowa State University

Ames, IA

- Designed and built User Equipment (UE) units for the ARA Wireless Living Lab, using Git for version control, C for firmware development, and Jupyter notebooks for testing, which enabled deployment of several units and enhanced wireless connectivity and affordability for rural and agricultural users.
- Reduced provisioning time for Linux-based User Equipment nodes by up to 70%, by authoring Ansible YAML playbooks that automated package installation and network configuration and standardized system images across a fleet of machines deployed in a distributed wireless testbed
- Configured Quectel antennae using Linux's Minicom to establish communication with the Iowa State University ARA internal network.

PROJECTS

Verbatim Speech Helper | *TypeScript, React, Python, SQLite, Railway*

August 2026

- Engineered a full-stack speech fluency app using TypeScript, FastAPI, and SQLite which deploys as 2 independently deployable services, exposing 8 REST endpoints for auth, content, and analytics.
- Eliminated over 90% of false-positive cursor jumps by developing a custom real-time LCS-based speech alignment algorithm that tracks natural speaking pace with fault tolerance, replacing a fixed-rate scrolling approach.
- Integrated an LLM API (Gemini) for on-demand content generation, capping usage to 5 requests/10 min per client to control costs on a public, unauthenticated endpoint.

Job Aggregator & Resume Matching | *Java, SpringBoot, PostgreSQL, Docker*

September 2026

- Built a Spring Boot service ingesting 50+ company job boards via scheduled REST polling of public Greenhouse and Lever APIs, normalizing data into PostgreSQL and Flyway-managed migrations.
- Developed a title-normalization pipeline that collapsed over 25% of duplicate listings across boards, using description similarity thresholds to consolidate reposts and duplicates.
- Implemented resume-to-posting matching using Gemini embeddings and cosine similarity to rank over 1000 postings by relevance.

MicroCART Quadcopter | *C, Python, OpenCV, Linux, RaspberryPi*

September 2025 - May 2026

- Designed and developed a custom semi-autonomous quadcopter integrating a Raspberry Pi and camera module with a Crazyflie UAV platform with optimized C-based firmware in a Linux virtual environment for real-time computer vision-based navigation.
- Achieved stable semi-autonomous flight with real-time vision-based navigation, which improved rate of safe landing by over 30% by developing C firmware that bridges a Raspberry Pi Zero to Crazyflie UAV over UART link and converts ArUco position data to setpoints for custom PID controller.
- Engineered a custom quadcopter frame upgrade to improve structural robustness and flight performance, supporting additional onboard computing hardware.

TuneFlow | *Java, Postman, Android Studio, Spotify WebAPI*

August - November 2024

- Increased daily active users by 40%, by developing an Android social-app frontend in Java that integrates the Spotify Web API via Volley for personalized recommendations.
- Designed multiple unique user interfaces, thoroughly tested using Postman, Mockito, and JUnit to increase user retention and returning users.

TECHNICAL SKILLS

Languages: Java, C, Python, TypeScript, SQL, HTML/CSS, Verilog, XML

Tools: Git, Linux, React, Ansible, LaTeX